Objective: Learn how to scan a network and identify open ports using Nmap.

1. Network Details

Local IP Address: 192.168.172.64

Subnet Mask: 255.255.255.0

Network Range: 192.168.172.0/24

2. Scanning Results

Basic Network Scan (nmap 192.168.172.0/24)

Active Devices Found:

192.168.172.79

Open Port: 53/tcp (DNS Service)

MAC Address: CE:A7:33:04:FC:78

Possible Device: Router or DNS Server

192.168.172.64 (Own Device)

Open Ports:

135/tcp → Microsoft RPC

139/tcp → NetBIOS

445/tcp → SMB (Windows File Sharing)

7070/tcp → RealServer (Streaming Service)

Service Detection Scan (nmap -sV 192.168.172.79)

All 1000 scanned ports were filtered (no response).

Possible Reasons: Firewall blocking requests or no exposed services.

Aggressive Scan (Optional) (nmap -A 192.168.172.1)

Host was unresponsive.

Likely Explanation: No device at this IP, or it’s blocking ping requests.

3. Analysis & Findings

Security Notes:

Windows file-sharing ports (139, 445) are open on 192.168.172.64 → These should be closed if not needed, as they can be exploited.

Router (192.168.172.79) does not expose services → Good security practice.

```
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
7070/tcp open  realserver

Nmap done: 256 IP addresses (2 hosts up) scanned in 10.84 seconds

E:\nmap>nmap -sV 192.168.172.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-29 22:14 India Standard Time
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.01 seconds

E:\nmap>nmap -A 192.168.172.1nmap -Pn -sV 192.168.172.79
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-29 22:16 India Standard Time
Failed to resolve "192.168.172.1nmap".
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 12.00% done; ETC: 22:20 (0:03:03 remaining)
^C
E:\nmap>nmap -A 192.168.172.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-29 22:18 India Standard Time
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.46 seconds

E:\nmap>nmap -Pn -sV 192.168.172.79
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-29 22:18 India Standard Time
Nmap scan report for 192.168.172.79
Host is up.
All 1000 scanned ports on 192.168.172.79 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 204.29 seconds

E:\nmap>nmap -Pn -sV 192.168.172.79
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-29 22:22 India Standard Time
```

```
E:\nmap>nmap 192.168.172.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-29 22:12 India Standard Time
Nmap scan report for 192.168.172.79
Host is up (0.0026s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
53/tcp open  domain
MAC Address: CE:A7:33:04:FC:78 (Unknown)

Nmap scan report for 192.168.172.64
Host is up (0.00062s latency).
Not shown: 996 closed tcp ports (reset)
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
7070/tcp open  realserver

Nmap done: 256 IP addresses (2 hosts up) scanned in 8.84 seconds

E:\nmap>nmap 192.168.172.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-29 22:14 India Standard Time
Nmap scan report for 192.168.172.79
Host is up (0.0045s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
53/tcp open  domain
MAC Address: CE:A7:33:04:FC:78 (Unknown)

Nmap scan report for 192.168.172.64
Host is up (0.00019s latency).
Not shown: 996 closed tcp ports (reset)
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
7070/tcp open  realserver

Nmap done: 256 IP addresses (2 hosts up) scanned in 10.84 seconds

E:\nmap>nmap -sV 192.168.172.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-29 22:14 India Standard Time
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
```